7 The area of a sector of a circle is $36.288\,\text{cm}^2$. The angle of the sector is θ radians and the radius of the circle is $r\,\text{cm}$.

(a) Find an expression for θ in terms of r. **[1]**

The perimeter of the sector is $24.48\,\text{cm}$.

(b) Show that $\theta = \dfrac{24.48}{r} - 2$. **[1]**

(c) Find the possible values of r. **[3]**

7(a)

7(b)

7(c)